



BohleBots Pompeii

Region: Germany, NRW, Haan League: 2 vs 2 LWL



<https://github.com/bohlebots-pompeii>



Method / Robot Production / Design

Materials:

- 16 Volt Lithium-Ion Akkus
- Pi-Camera 3
- ESP 32 Wroom 32
- Lightgate
- Omniwheels
- Dribbler + 1 motor (brushless)
- Kicker
- line-sensor-ring
- 4 Motors (brushless)
- Mirror
- CM5
- Hailo8L
- 3D print of PLA

Software:

- We use AI for tracking the ball and the two goals
- We use a yolo8 for vision processing



Generall informations:

- We used Fusion 360 to design our robots
- We designed our custom circuit boards in KiCAD and they were manufactured by JLCPCB
- The programm is written in C++ and camera code in Python with CLion and PyCharm
- The whole rest of the robots is build of 3D-prints
- The price of one robot is 821,70€

Logo of the Team



Team Photo



Abstract

We are Team Pompeii of the Bohlebots. We are coached by Roland Stiebel from Städtisches Gymnasium Haan in North Rhine-Westphalia, Germany. Our team consists of five members. Julius Gerhardus and Timo Markgraf worked on the software. Constantin Wiechert, Joel Assmann, and Jonathan Weiss are responsible for the hardware. Four of us are in eleventh grade, and one is in tenth grade. Nevertheless, we all work together on equal terms. This combination of experience and fresh ideas makes our team spirit particularly strong. Our team brings together different strengths. This clear division of responsibilities allows us to collaborate efficiently. In the software area, we develop the logic for our robots. We process sensor data in real time, calculate positions, and adjust driving and shooting decisions. Visual perception is also important. Using a camera and a mirror system, we are able to see everything around us. We use artificial intelligence to detect the ball and the goal. Another focus is the optimization of our system architecture to ensure that both robots work well together. In the hardware area, we concentrate on construction and mechanical efficiency. We inspect and improve the chassis, the drive system, and the ball-handling mechanism. Our goal is to build a robust and high-performance system that also functions reliably in competitions. Weight distribution, traction, and durability are important to us. We test components, replace worn parts, and improve the mechanics. We also try out new design approaches and test them directly during gameplay. Teamwork also plays an important role. We support each other and learn from one another. Through joint planning and testing, we create a fully functional overall system. As Bohlebots, we stand for innovation, unity, and enthusiasm for technology. We are proud to represent our school and look forward to further developing our skills and taking on new challenges.

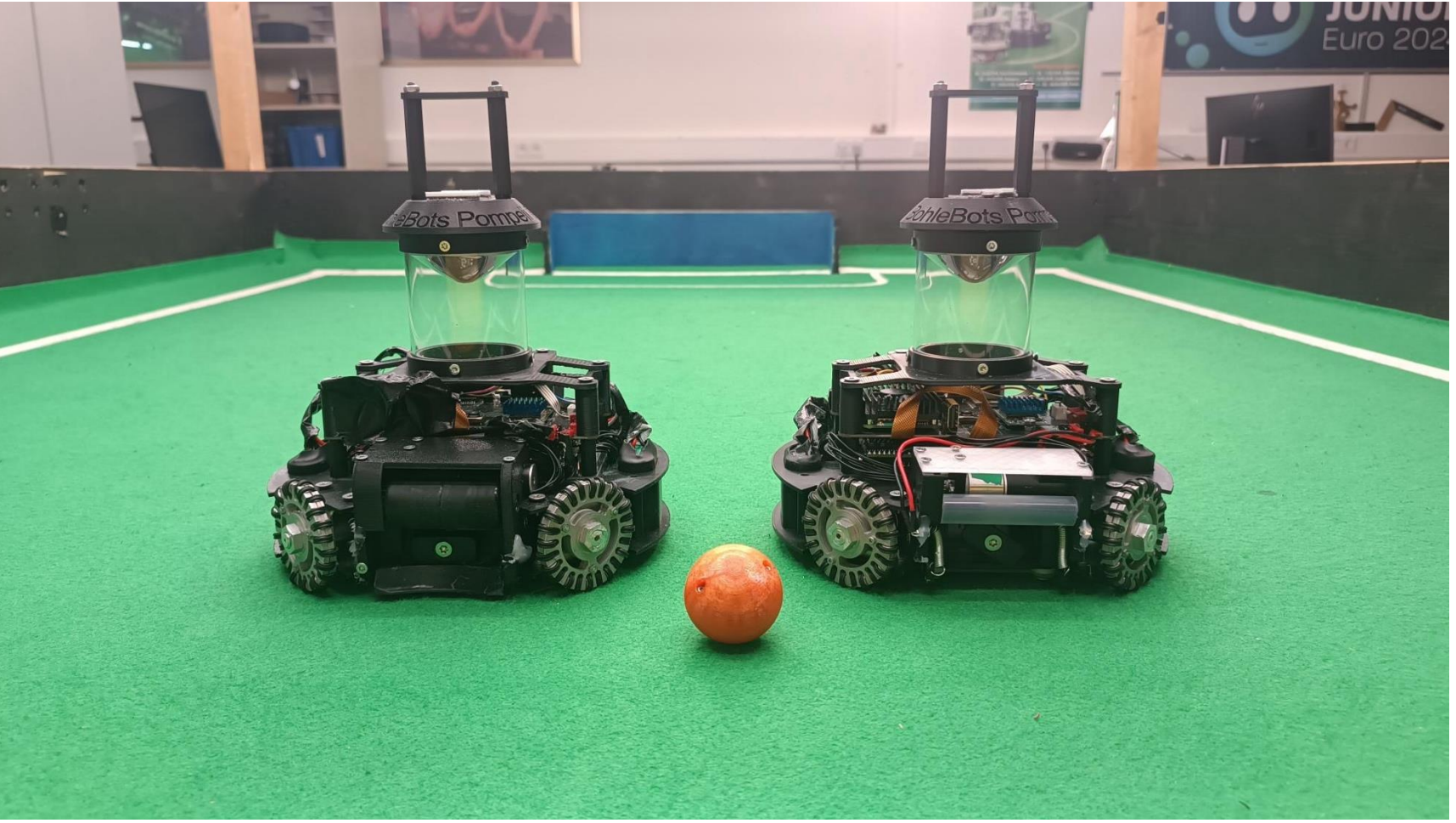
Data / Results / Discussion

During development, the robot went through several iterative hardware and software revisions driven by testing on the field. The final system is built around a two-PCB architecture: a lower board handling motor drivers, and a dedicated ESP32 for low-level control, and an upper board with a second ESP32 and a CM5 module with a Hailo-8L Ai accelerator for vision processing. This separation improved real-time responsiveness while allowing more complex perception algorithms to run in parallel.

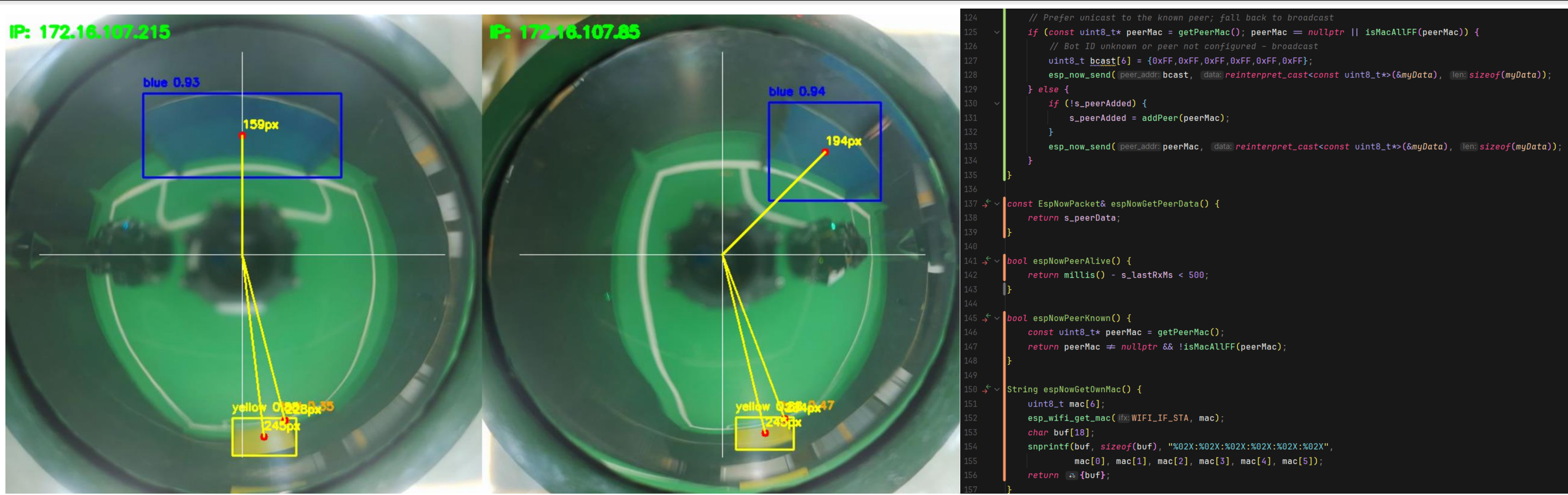
Sensor data from the omnidirectional camera, line sensors was used to estimate position and orientation. Early tests showed strong noise and drift, especially during fast movements, which we solved by implementing better AI models. After tuning, position stability and rotational accuracy improved noticeably, resulting in more reliable navigation and ball tracking.

On the software side, control behaviors such as ball capture, striking, and goalie logic were refined through repeated testing. PID controllers were adjusted based on logged movement data to reduce overshoot and oscillations. Overall, systematic testing and data-driven modifications led to a more robust, accurate, and competition-ready robot.

Robot Photo



Fusion Design of one robot



AI and Code picture